

Measure Theory

Functional Analysis

Morales Carmona Ronny

Autumn 2026

Contents

Morales C. Ronny

The Lebesgue Outer Measure

Measurable Spaces

Measurable Functions

Measure Spaces

The Lebesgue Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class Theorem

Product Spaces

Fubini's Theorem

Extension to Complex-Valued Functions

Lebesgue Integration on $\mathbb{R}^n$

Definition

The set $\overline{\mathbb{R}}$ of **extended real numbers** is defined as

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}.$$

In $\overline{\mathbb{R}}$, the two (new) elements $+\infty, -\infty$ play the role of a number in the following sense:

- a) For any number $x \in \mathbb{R}$, $x + \infty = \infty$ and $x - \infty = -\infty$;
- b) $x \cdot \infty = \infty$ for $x \in \mathbb{R}^+$ and $x \cdot \infty = -\infty$ for $x \in \mathbb{R}^-$;
- c) $\infty + \infty = \infty$, $-\infty - \infty = -\infty$;
- d) $\infty \cdot \infty = \infty$ and $-\infty \cdot \infty = -\infty$.

We also declare $-\infty < x < \infty$ for all $x \in \mathbb{R}$.

The set

$$\overline{\mathbb{R}}_+ = \mathbb{R}^+ \cup \{\infty\} = (0, \infty]$$

is called the **extended positive real numbers**.

Definition

We say that an interval in $\mathbb{R}$ is **degenerate** [respectively, **nondegenerate**] if it contains at most one point [respectively, if it contains at least two points]. Two intervals are disjoint if their intersection is empty, similarly, we say that two intervals are **nonoverlapping** if their intersection is degenerate.

Definition

The **length** $\ell(I)$ of an open bounded interval $I \doteq (a, b)$ with $a \leq b^\dagger$ is defined by

$$\ell(I) = b - a.$$

Similarly, for a closed bounded interval $J \doteq [a, b]$ we define

$$\ell(J) = b - a.$$

For any unbounded interval we define its length to be ∞ .

[†]We come to the convention of understanding $(a, b) = \emptyset$ in the case $a = b$.

The Lebesgue Outer Measure
Measurable Spaces
Measurable Functions
Measure Spaces
The Lebesgue Measure
Simple Functions
Abstract Integration
Properties of $\int d\mu$
Cross Sections
Monotone Class Theorem
Product Spaces
Fubini's Theorem
Extension to Complex-Valued Functions
Lebesgue Integration on $\mathbb{R}^n$

Definition

*Let $A \subseteq \mathbb{R}$. A collection $\mathcal{C}$ of open subsets of $\mathbb{R}$ is called an **open cover** of A if A is contained in the union of all the sets in $\mathcal{C}$.*

Definition

Let $\mathcal{C}$ be the class of all countable open covers $I \doteq \{I_n\}_{n \in \mathbb{N}}$ of A . The **Lebesgue outer measure** (or simply, **outer measure**) $m^*(A)$ of a set $A \subseteq \mathbb{R}$ is defined by

$$m^*(A) = \inf_{I \in \mathcal{C}} \left\{ \sum_{n \in \mathbb{N}} \ell(I_n) \right\}.$$

Theorem 1.1

Every countable subset of $\mathbb{R}$ has Lebesgue outer measure 0.

Definition

If $t \in \mathbb{R}$ and $A \subseteq \mathbb{R}$, then the translation $t + A \equiv A + t$ is defined by

$$t + A = \{t + a : a \in A\}.$$

Theorem 1.2 (Properties of Outer Measure)

Suppose A and B are subsets of $\mathbb{R}$. Then

- i) **(Monotonicity)** if $A \subset B$, then $m^*(A) \leq m^*(B)$;
- ii) **(Translation invariance)** if $t \in \mathbb{R}$, then $m^*(A + t) = m^*(A)$;
- iii) **(Homogeneity)** if $t > 0$, then $m^*(tE) = tm^*(E)$;
- iv) **(σ -subadditivity)** if (A_n) is a sequence of subsets of $\mathbb{R}$, then

$$m^*\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} m^*(A_n).$$

Definition

Let $A \subseteq \mathbb{R}$. An open cover $\mathcal{C}$ of A is said to have a **finite subcover** if A is contained in the union of some finite list of sets in $\mathcal{C}$.

Example 1.3

The collection $\{(k, k + 2) : k \in \mathbb{Z}^+\}$ is an open cover of $[2, 5]$. This open cover has a finite subcover because $[2, 5] \subset (1, 3) \cup (2, 4) \cup (3, 5) \cup (4, 6)$.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
Functions

Lebesgue
Integration on $\mathbb{R}^n$

Definition

*A subset $A \subseteq \mathbb{R}$ is said to be **compact** if every open cover has a finite subcover.*

Theorem 1.4

The closed and bounded subsets of $\mathbb{R}$ are compact.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Theorem 1.5 (Heine-Borel Theorem)

A subset A of $\mathbb{R}$ is compact if and only if it is closed and bounded.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$ **Theorem 1.6**

Let $a, b \in \mathbb{R}$, with $a \leq b$. Then $m^([a, b]) = b - a$.*

Theorem 1.7

Every interval in $\mathbb{R}$ that contains at least two distinct elements is uncountable.

Lemma 1.8

The relation $\sim$ on $\mathbb{R}$ defined by $x \sim y$ if and only if $x - y \in \mathbb{Q}$ is an equivalence relation. In particular, it is an equivalence relation on $[-1, 1]$.

Definition

Let $\sim$ be the relation defined as in the previous lemma. Then by using the axiom of choice, there exists a set V such that $V \cap [a]$ contains exactly one element for each $a \in [-1, 1]$. Any such set is called a **Vitali set**.

Theorem 1.9

There exist disjoint subsets A and B of $\mathbb{R}$ such that

$$m^*(A \cup B) \neq m^*(A) + m^*(B).$$

Theorem 1.10

There does not exist a function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ satisfying all the following properties:

- i) $\mu(I) = \ell(I)$ for every open interval I of $\mathbb{R}$.
- ii) $\mu\left(\bigsqcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n)$.
- iii) $\mu(A + t) = \mu(A)$ for all $A \subset \mathbb{R}$ and every $t \in \mathbb{R}$.

Definition

Given a set X , a **σ -algebra** $\mathcal{M}$ on X is a collection of subsets of X , called the **$\mathcal{M}$ -measurable subsets of X** , satisfying the following conditions:

- Meas 1)** X is an element of $\mathcal{M}$.
- Meas 2)** $\mathcal{M}$ is closed under complements.
- Meas 3)** $\mathcal{M}$ is closed under denumerable unions.

The pair $(X, \mathcal{M})$ is known as a **measurable space**. When both X and its σ -algebra are understood, we will simply refer to the elements in $\mathcal{M}$ as **measurable subsets**.

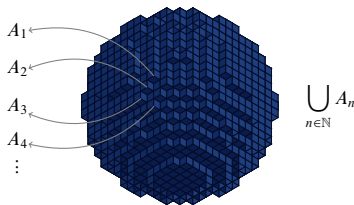


Figure 1: Geometrical meaning of **Meas 3**

Contents

Morales C. Ronny

The Lebesgue Outer Measure

Measurable Spaces

Measurable Functions

Measure Spaces

The Lebesgue Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class Theorem

Product Spaces

Fubini's Theorem

Extension to Complex-Valued Functions

Lebesgue Integration on $\mathbb{R}^n$

Remark

Let us give a few easy consequences of the definition:

1. $\emptyset \in \mathcal{M}$.
2. If $A_n \in \mathcal{M}$ for every $n \in \mathbb{N}$, one has also $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{M}$.
3. Since we may take $A_n = \emptyset$ for all $n > n_0$, property (iii) implies that $\mathcal{M}$ is closed under finite unions, meaning that the union of finitely many elements of $\mathcal{M}$ is in $\mathcal{M}$, and then $\mathcal{M}$ is also closed under finite intersections using (ii).

Remark

Given any set X , the collection $\mathcal{M} = \{\emptyset, X\}$ is obviously a σ -algebra, called the **trivial σ -algebra**. This is the smallest[†] σ -algebra possible. By contrast, the largest σ -algebra is the power set $\mathcal{P}(X)$. All interesting examples fall somewhere between these two extremes.

[†]We define the “smaller than” relation $\leq$ on sets as $A \leq B$ if and only if $A \subseteq B$, and by contrast, we define the relation $\geq$ “larger than” as $P \geq Q$ if and only if $Q \leq P$.

Remark

The intersection of any two σ -algebras $\mathcal{M} \cap \mathcal{M}'$ is another σ -algebra, we can extend the statement to the intersection of an arbitrary family of σ -algebras. Therefore, given any collection of subsets $\mathcal{A} \subset \mathcal{P}(X)$, there is a unique smallest σ -algebra $\mathcal{S} \supseteq \mathcal{A}$, namely, the intersection of all σ -algebras that contain $\mathcal{A}$.

Definition

Let $\mathcal{A}$ be a collection of subsets of X . The **σ -algebra generated by $\mathcal{A}$** , denoted by $\sigma(\mathcal{A})$, is the σ -algebra defined as

$$\sigma(\mathcal{A}) = \bigcap \{ \mathcal{M} \subseteq \mathcal{P}(X) : \mathcal{M} \text{ is a } \sigma\text{-algebra and } \mathcal{M} \supset \mathcal{A} \}.$$

Definition

For a topological space $(X, \mathcal{T})$, the σ -algebra $\sigma(\mathcal{T})$ generated by the open sets is called the **Borel σ -algebra** on X and is denoted by $\mathcal{B}(X)$. The elements of $\mathcal{B}(X)$ are called the **Borel sets** on X .

Example 1.11

Consider $\mathbb{R}$ with its usual topology. Prove that

- a) the Borel σ -algebra is generated by (*) the infinite left-open intervals (a, ∞) , and (**) the closed intervals $[a, b]$;
- b) all singletons $\{a\}$ with $a \in \mathbb{R}$ are Borel sets on $\mathbb{R}$.

Definition

Given two measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$, a map $f : X \rightarrow Y$ is said to be a **measurable function** (or an **$\mathcal{M}$ -measurable function in the $\mathcal{N}$ sense**) if the inverse image of every measurable set is measurable:

$$f^{-1}(\mathcal{N}) \subseteq \mathcal{M}.$$

Theorem 1.12

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are measurable functions between measurable spaces, then the composition $g \circ f : X \rightarrow Z$ is a measurable function.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$ **Theorem 1.13**

If X and Y are topological spaces, then every continuous function $f : X \rightarrow Y$ is Borel measurable in the Borel sense.

Theorem 1.14

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be two measurable spaces, and consider a function $f : X \rightarrow Y$. A sufficient condition for f to be measurable is that there exists a subclass $\mathcal{C}$ of $\mathcal{N}$ such that $\sigma(\mathcal{C}) = \mathcal{N}$ and $f^{-1}(\mathcal{C}) \subseteq \mathcal{M}$.

Remark

Let f be as in the previous definition, let $Y = \mathbb{R}$ and let $\mathcal{N} = \mathcal{B}(\mathbb{R})$. Appealing to the previous theorem and the previous example, it suffices to prove that the sets $f^{-1}((a, b))$ are $\mathcal{M}$ -measurable sets for every $a < b$. It is even enough to consider the sets $f^{-1}((-\infty, a))$, $f^{-1}((a, \infty))$ or $f^{-1}([a, b])$.

Theorem 1.15

Every increasing function defined on a Borel set is Borel measurable.

Definition

If X is a set and A is a subset thereof, then we define the **characteristic function** of A , by

$$\mathbf{1}_A(x) \equiv \chi_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Example 1.16

Show that if $(X, \mathcal{M})$ is a measurable space then the characteristic function $\chi_A : X \rightarrow \overline{\mathbb{R}}$ of a subset $A \subset X$ is measurable if and only if $A \in \mathcal{M}$. Show further that for any $a \in \mathbb{R}$ we have $\{x \in X : f(x) = a\} \in \mathcal{M}$.

Definition

If $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are two measurable spaces then we define the **product measurable space** $(X \times Y, \mathcal{M} \otimes \mathcal{N})$, by setting the σ -algebra $\mathcal{M} \otimes \mathcal{N}$ on $X \times Y$ to be the σ -algebra generated by all sets of the form $A \times B$ where $A \in \mathcal{M}$ and $B \in \mathcal{N}$, in symbols,

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\mathcal{C}), \quad \mathcal{C} = \{A \times B : A \in \mathcal{M}, B \in \mathcal{N}\}.$$

Given any measurable space $(X, \mathcal{M})$, the product sigma algebra on the n -fold X^n , denoted by $\mathcal{M}^{\otimes n}$, is defined as $\mathcal{M}^{\otimes n} = \mathcal{M} \otimes \mathcal{M}^{\otimes(n-1)}$, with $\mathcal{M}^{\otimes 1} \equiv \mathcal{M}$.

Lemma 1.17

Let $(X, \mathcal{M})$, $(Y_1, \mathcal{N}_1)$, and $(Y_2, \mathcal{N}_2)$ be measurable spaces, and equip the product $Y_1 \times Y_2$ with the product σ -algebra $\mathcal{N}_1 \otimes \mathcal{N}_2$. Let $f_j : X \rightarrow Y_j$ and define $f : X \rightarrow Y_1 \times Y_2$ by setting $f = (f_1, f_2)$. Then f is measurable if and only if f_1 and f_2 are both measurable.

Theorem 1.18

$$\mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R})^{\otimes n}.$$

Theorem 1.19

Let $(X, \mathcal{M})$ be a measurable space, and let f and g be measurable functions from X into $\mathbb{R}$. Then the functions $f + g$, fg , and f/g are all measurable (f/g provided that $g(x) \neq 0$ for $x \in X$).

Definition

Take an arbitrary function $f : X \rightarrow \mathbb{R}$, consider the sets

$$P = \{x \in X : f(x) \geq 0\} \quad \text{and} \quad N = \{x \in X : f(x) \leq 0\}.$$

We define the **positive part** and the **negative part** of f as

$$f^+ = f\chi_P \quad \text{and} \quad f^- = -f\chi_N,$$

respectively.

Remark

The positive and negative parts of a function can be written as

$$f^+ = \sup(f, 0) \quad \text{and} \quad f^- = \sup(-f, 0),$$

respectively. Moreover, for any $f : X \rightarrow \mathbb{R}$ we can express f and $|f|$ as

$$f = f^+ - f^- \quad \text{and} \quad |f| = f^+ + f^-.$$

Theorem 1.20

For a function $f : X \rightarrow \mathbb{R}$, the following are equivalent:

- i) f is a measurable function.*
- ii) f^+ and f^- are both measurable functions.*
- iii) $f^{-1}([0, \infty))$ and $f^{-1}((-\infty, 0])$ are measurable sets and $|f|$ is a measurable function.*

Definition

Let $(X, \mathcal{M})$ be a measurable space. A function $f : X \rightarrow [-\infty, \infty]$ is called an **extended measurable map** (or simply measurable, when the target space is understood) if

- i) $f^{-1}(\mathcal{B}(\mathbb{R})) \subseteq \mathcal{M}$.
- ii) $f^{-1}(\{-\infty\}) \in \mathcal{M}$ and $f^{-1}(\{\infty\}) \in \mathcal{M}$.

Theorem 1.21

Suppose $(X, \mathcal{M})$ is a measurable space and let $f : X \rightarrow \overline{\mathbb{R}}$ be a function with the property that $f^{-1}((a, \infty])$ is measurable for all $a \in \mathbb{R}$. Then f is an $\mathcal{M}$ -measurable function.

Theorem 1.22

Let (f_n) be a sequence of measurable functions from E into $\overline{\mathbb{R}}$. Then,

$$\sup_{n \in \mathbb{N}} f_n, \quad \inf_{n \in \mathbb{N}} f_n, \quad \limsup f_n, \quad \text{and} \quad \liminf f_n$$

are also measurable functions. In particular, if the sequence (f_n) converges pointwise, its limit $\lim f_n$ is a measurable function.

Definition

Given a measurable space $(X, \mathcal{M})$, a **positive measure**[†] μ on X is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ satisfying:

Meas 4) $\mu(\emptyset) = 0$;

Meas 5) (**σ -additivity**)

$$\mu \left(\bigcup_{n \in \mathbb{N}} A_n \right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

If the series on the right-hand of the previous equation is not convergent, it is given the value ∞ . The triple $(X, \mathcal{M}, \mu)$ is called a **measure space**. The quantity $\mu(X)$ is called the **total mass** of the measure μ . Given any $A \in \mathcal{M}$, the (extended) number $\mu(A)$ is called the **measure** of A . A measurable subset of X is called a **μ -null set** if it has measure zero.

[†]There are constructions of signed measures and complex measures, but these constructions are out of the scope of this course, so we will simply refer to a positive measure as a measure.

Definition

Let X be a nonempty set and let I be an index-set. A family $\{X_i\}_{i \in I}$ of subsets $X_i \subseteq X$ of X is called a **partition** of X if

$$X_i \cap X_j = \emptyset \quad \text{for } i \neq j, \quad \text{and} \quad \bigcup_{i \in I} X_i = X.$$

Definition

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then we say that the measure μ is

- **finite** if its total mass is finite.
- **probability measure** if it has unit mass. In this case, we say that $(X, \mathcal{M}, \mu)$ is a **probability space**.
- **σ -finite** if there exists a partition (A_n) of X of measurable sets such that $\mu(A_n) < \infty$ for each n .
- **equal** to another measure ν on X , if $\mu(A) = \nu(A)$ for each $A \in \mathcal{M}$.

Theorem 1.23 (Properties of a Measure)

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $A, B, A_n \in \mathcal{M}$ for all $n \in \mathbb{N}$. Then

- i) **(Monotonicity)** If $A \leq B$, then $\mu(A) \leq \mu(B)$. If in addition $\mu(B) < \infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

- ii) **(Continuity from below)** [respectively, **from above**] If (A_n) is an increasing [respectively, decreasing with $\mu(A_1) < \infty$] sequence, then

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \lim_{n \rightarrow \infty} \uparrow \mu(A_n), \left[\text{respectively, } \mu\left(\bigcap_{n \in \mathbb{N}} A_n\right) = \lim_{n \rightarrow \infty} \downarrow \mu(A_n) \right].$$

- iii) **(σ -subadditivity)** $\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} \mu(A_n).$

Theorem 1.24 (Egoroff's Theorem)

Suppose $(X, \mathcal{M}, \mu)$ is a measure space and that μ is finite. Suppose (f_n) is a sequence of measurable functions $f_n : X \rightarrow \mathbb{R}$ that converges pointwise to $f : X \rightarrow \mathbb{R}$. Then for every $\varepsilon > 0$, there exists a set $E \in \mathcal{M}$ such that $\mu(X \setminus E) < \varepsilon$ and (f_n) converges uniformly to f on E .

Definition

If μ is a positive measure on $(X, \mathcal{M})$, and $C \in \mathcal{M}$, the **restriction of μ to C** is denoted by $\mu|_C$ and defined as

$$\mu|_C(A) = \mu(A \cap C)$$

for all $A \in \mathcal{M}$.

Example 1.25

The following are some examples of measures.

- Let a be any fixed point of X , and set

$$\delta_a(A) = \begin{cases} 1 & \text{if } a \in A, \\ 0 & \text{if } a \notin A, \end{cases}$$

for each measurable subset A . Then function $\delta_a(A)$ defines a measure on X , known as the **Dirac measure**.

- In the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$, we define the **counting measure** by

$$\mu(A) := \text{card}(A).$$

- The restriction of the Lebesgue outer measure to $\mathcal{B}(\mathbb{R})$ (the Borel σ -algebra on $\mathbb{R}$) defines a measure on $\mathbb{R}$.

Definition

Given a measure space $(X, \mathcal{M}, \mu)$, we say that a property $\mathcal{P}(x)$ depending on x , holds **μ -almost everywhere** (abbreviated μ -a.e., or simply a.e. when the measure is understood from the context) if $\{x \in X : \mathcal{P}(x) \text{ holds}\} \in \mathcal{M}$ and

$$\mu(\{x \in X : \mathcal{P}(x) \text{ does not hold}\}) = 0.$$

Definition

A measure space $(X, \mathcal{M}, \mu)$ is said to be **complete** if $\mathcal{M}$ contains all subsets of μ -null sets, i.e., if $A \in \mathcal{M}$, $\mu(A) = 0$ and $B \subseteq A$ imply that $B \in \mathcal{M}$.

Theorem 1.26 (Completion of a measure space)

Given any measure space $(X, \mathcal{M}, \mu)$, let $\mathcal{N}$ be the class of all μ -null measurable subsets of X , and define $\widehat{\mathcal{M}}^\mu$ as

$$\widehat{\mathcal{M}}^\mu = \{E \cup A : E \in \mathcal{M} \text{ and } A \subseteq N \text{ for some } N \in \mathcal{N}\},$$

and define $\widehat{\mu} : \widehat{\mathcal{M}}^\mu \rightarrow [0, \infty]$ by $\widehat{\mu}(\widehat{E}) = \mu(E)$ for each $\widehat{E} \in \widehat{\mathcal{M}}^\mu$. Then the following assertions hold true

- i) $\widehat{\mu}$ is well-defined.
- ii) $\widehat{\mathcal{M}}^\mu$ is a σ -algebra on X and $\mathcal{M} \subseteq \widehat{\mathcal{M}}^\mu$.
- iii) $\widehat{\mu}$ is a measure on $\widehat{\mathcal{M}}^\mu$ such that $\widehat{\mu}(E) = \mu(E)$ for every $E \in \mathcal{M}$.
- iv) $(X, \widehat{\mathcal{M}}^\mu, \widehat{\mu})$ is a complete measure space, usually referred to as the **completion** of the measure space $(X, \mathcal{M}, \mu)$.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Remark

With the preceding results at hand, we are now in position to construct the Lebesgue measure and the Lebesgue σ -algebra.

Theorem 1.27

Suppose A and G are disjoint subsets of $\mathbb{R}$ and G is open. Then

$$m^*(A \cup G) = m^*(A) + m^*(G).$$

Theorem 1.28

Suppose A and F are disjoint subsets of $\mathbb{R}$ and F is closed. Then

$$m^*(A \cup F) = m^*(A) + m^*(F).$$

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$

Theorem 1.29

Suppose $B \subseteq \mathbb{R}$ is a Borel set. Then for every $\varepsilon > 0$, there exists a closed set $F \subseteq B$ such that $m^(B \setminus F) < \varepsilon$.*

Theorem 1.30

Suppose A and B are disjoint subsets of $\mathbb{R}$ and $B \in \mathcal{B}$. Then

$$m^*(A \cup B) = m^*(A) + m^*(B).$$

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$

Theorem 1.31

There exists a set $B \subseteq \mathbb{R}$ with $m^(B) < \infty$ such that $B \notin \mathcal{B}$.*

Definition

The **Lebesgue measure** $\lambda : \mathcal{B} \rightarrow \overline{\mathbb{R}}$ on $(\mathbb{R}, \mathcal{B})$ is defined as

$$\lambda(B) = m^*(B)$$

for all $B \in \mathcal{B}$.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Proposition 1.32

$(\mathbb{R}, \mathcal{B}, \lambda)$ is a measure space.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$

Definition

The **Lebesgue σ -algebra** $\mathcal{L}(\mathbb{R})$ (or simply, $\mathcal{L}$) is the completion of the Borel σ -algebra with respect to the Lebesgue outer measure. Any set in $\mathcal{L}$ is called a **Lebesgue set**.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$

“Passing from Borel to Lebesgue measurable functions is the work of the devil. Don’t even consider it!”

- Barry Simon (winner of the American Mathematical Society Steele Prize for Lifetime Achievement)

Definition

A **simple function** is a measurable function $h : X \rightarrow \overline{\mathbb{R}}$ that takes on only a finite set of extended values $a_1, a_2, \dots, a_n$. The set of all nonnegative simple functions is denoted by $\mathcal{E}_+$.

Remark

Let f be a simple function taking values $a_1, a_2, \dots, a_n$. Since $A_i = f^{-1}(\{a_i\})$ is a measurable subset of X for each a_i , we can write a simple function as a linear combination of characteristic functions[†]

$$f = a_j \chi_{A_j} \quad \text{where} \quad a_i \neq 0 \quad \forall \quad j \in \{1, \dots, n\}.$$

The latter is known as the **canonical representation** of f .

[†]Henceforth, we will use the summation convention.

[The Lebesgue Outer Measure](#)[Measurable Spaces](#)[Measurable Functions](#)[Measure Spaces](#)[The Lebesgue Measure](#)[Simple Functions](#)[Abstract Integration](#)[Properties of \$\int d\mu\$](#) [Cross Sections](#)[Monotone Class Theorem](#)[Product Spaces](#)[Fubini's Theorem](#)[Extension to Complex-Valued Functions](#)[Lebesgue Integration on \$\mathbb{R}^n\$](#)

Lemma 1.33

Let f be a nonnegative measurable function on X . Then there exists an increasing sequence (f_n) of nonnegative simple functions such that $f_n \rightarrow f$ pointwise as $n \rightarrow \infty$. If f is also bounded, the sequence (f_n) can be chosen to converge uniformly to f .

Theorem 1.34 (Approximation Lemma)

Any measurable function $f : X \rightarrow \mathbb{R}$ is the limit of a sequence of simple functions. The sequence can be chosen to be monotone increasing at all positive values of f , and monotone decreasing at negative values.

Remark

Throughout this section, we will consider a complete measure space $(X, \mathcal{M}, \mu)$.

Definition

Let $f \in \mathcal{E}_+$ with canonical representation $f = \alpha_j \chi_{A_j}$. Then the **integral of f with respect to μ** is defined by

$$\int d\mu f := \alpha_j \mu(A_j).$$

We will adopt the convention that $0 \cdot \infty$ and $\infty \cdot 0$ should both be interpreted to be 0.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Theorem 1.35

$\int d\mu f$ is well-defined provided that $f \in \mathcal{E}_+$.

Definition

Let $f : X \rightarrow [0, \infty]$ be a measurable function. The **integral of f with respect to μ** is defined by

$$\int d\mu f := \sup_{h \in \Phi_f^+} \int d\mu h, \quad \Phi_f^+ = \{h : \mathcal{E}^+ \ni h \leq f\}.$$

The symbols

$$\int d\mu(x) f(x), \quad \int \mu(dx) f(x)$$

are sometimes used to denote the value of the integral. If the measure is understood, we will simply write

$$\int f.$$

Theorem 1.36

Let $f, g : X \rightarrow [0, \infty]$ and suppose that $f \leq g$. Then

$$\int f \leq \int g.$$

Theorem 1.37 (Beppo Levi's Monotone Convergence Theorem)

Let $(f_n) \subset \mathcal{E}^+$ be such that $f_n \nearrow f$, then

$$\int f = \lim \int f_n.$$

Definition

Let $f : X \rightarrow \mathbb{R}$ be a measurable function. Then we say that f is **integrable with respect to μ** (or **μ -integrable**) if

$$\int d\mu f^+ < \infty \quad \text{and} \quad \int d\mu f^- < \infty.$$

In this case, we define the **integral of f with respect to μ** (or simply, the **μ -integral of f**) to be the number

$$\int d\mu f = \int d\mu f^+ - \int d\mu f^-.$$

When the measure μ is clear from the context, we will simply write $\int f$ to denote the value of the integral.

Definition

Let $f : X \rightarrow [-\infty, \infty]$ be an extended measurable map.

- If $f(x) \in [0, \infty]$ for all $x \in X$, then we define

$$\int d\mu f = \begin{cases} \int d\mu \chi_{\{x: f(x) < \infty\}} f & \mu(\{x : f(x) = \infty\}) = 0, \\ \infty & \mu(\{x : f(x) = \infty\}) > 0. \end{cases}$$

- If either $\int f^+ < \infty$ or $\int f^- < \infty$, then we define

$$\int f = \int f^+ - \int f^-.$$

Theorem 1.38 (Properties of the Lebesgue Integral)

Let $f, g : X \rightarrow \mathbb{R}$ be measurable and μ -integrable functions, then

- i) **(Linearity)** $\int (f + bg) = \int f + b \int g$ for any $b \in \mathbb{R}$.
- ii) **(Monotonicity)** If $f \leq g$ a.e., then $\int f \leq \int g$.
- iii) **(σ -Additivity)** If $\{A_i\}_{i=1}^n$ is a partition of X , then

$$\int_{\bigcup A_i} f = \sum_{i=1}^n \int_{A_i} f.$$

- iv) **(Semi-nondegeneracy)** if $f = 0$ a.e., then $\int f = 0$, conversely, if $f \geq 0$ and $\int f = 0$, then $f = 0$ a.e.
- v) **(Absolute integrability)** f is integrable if and only if $|f|$ is integrable.
- vi) **(Triangle inequality for integrals)** $\left| \int f \right| \leq \int |f|$.

Theorem 1.39 (Lebesgue's Integrability Criterion)

If $f = g$ a.e. on X and if f is integrable on X , then g is integrable on X and

$$\int f = \int g.$$

The Lebesgue Outer Measure	
Measurable Spaces	
Measurable Functions	
Measure Spaces	
The Lebesgue Measure	
Simple Functions	
Abstract Integration	
Properties of $\int d\mu$	
Cross Sections	
Monotone Class Theorem	
Product Spaces	
Fubini's Theorem	
Extension to Complex-Valued Functions	
Lebesgue Integration on $\mathbb{R}^n$	

Definition

The space of all μ -integrable functions $f : X \rightarrow \mathbb{R}$ is denoted by $\mathcal{L}^{1,\mu}(X, \mathcal{M})$ or $\mathcal{L}^1(X, \mathcal{M}, \mu)$. We will also use the notation $\mathcal{L}_+^1(X, \mathcal{M}, \mu)$ for the space of all nonnegative μ -integrable functions. The name “space” is justified in the following theorem.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Theorem 1.40

$\mathcal{L}^1(X, \mathcal{M}, \mu)$ is a linear space.

Definition

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be two measurable spaces and let $\varphi : X \rightarrow Y$ be a measurable function. Let μ be a positive measure on $(X, \mathcal{M})$. We define the **pushforward** of μ under φ , denoted by $\varphi_*\mu$, via the formula

$$\varphi_*\mu(B) = \mu(\varphi^{-1}(B)), \quad \forall B \in \mathcal{N}.$$

The pushforward of μ under φ defines a positive measure on $(Y, \mathcal{N})$.

Theorem 1.41 (Change of Variables Formula)

Let $(Y, \mathcal{N})$ be a measurable space, let and let $\varphi : X \rightarrow Y$ be a measurable function. Then for any measurable function h on F , we have

$$\int_X d\mu \, h \circ \varphi = \int_Y d\varphi_* \mu \, h.$$

Theorem 1.42 (Riemann and Lebesgue Integral Connection)

Let f be a bounded real-valued function on $[a, b]$ and let λ be the Lebesgue measure on $\mathbb{R}$. Then the following statement hold true:

- i) $f \in \mathcal{R}[a, b]$ if and only if f is continuous λ a.e.
- ii) If $f \in \mathcal{R}[a, b]$, then $f \in \mathcal{L}[a, b]$, and

$$\int_a^b f = \int_{[a,b]} d\lambda f.$$

Theorem 1.43 (Fatou's Lemma)

Let (f_n) be a sequence of nonnegative measurable functions on X . Then

$$\int \liminf f_n \leq \liminf \int f_n$$

Theorem 1.44 (Lebesgue's Dominated Convergence)

Let (f_n) be a sequence in $\mathcal{L}^1(X, \mathcal{M}, \mu)$. We assume that there exists a measurable function f with values in $\mathbb{R}$ such that $f_n \rightarrow f$ a.e. and also that there exists a nonnegative measurable function g such that $\int d\mu g < \infty$ and for every $n \in \mathbb{N}$, $|f_n| \leq g$ a.e. Then $f \in \mathcal{L}^1(X, \mathcal{M}, \mu)$ and we have

$$\lim \int d\mu f_n = \int d\mu f.$$

Definition

Suppose X and Y are sets and $E \subset X \times Y$. Then for $a \in X$ and $b \in Y$, the **cross sections** $[E]_a$ and $[E]_b$ are defined by

$$[E]_a = \{y \in Y : (a, y) \in E\} \quad \text{and} \quad [E]_b = \{x \in X : (x, b) \in E\}.$$

Theorem 1.45

Suppose $\mathcal{M}$ is a σ -algebra on X and $\mathcal{N}$ is a σ -algebra on Y . If $E \in \mathcal{M} \otimes \mathcal{N}$, then

$$[E]_a \in \mathcal{M} \text{ for every } a \in X \quad \text{and} \quad [E]^b \in \mathcal{N} \text{ for every } b \in Y.$$

Definition

Suppose X and Y are sets and $f : X \times Y \rightarrow \mathbb{R}$ is a function. Then for $a \in X$ and $b \in Y$, the cross section functions $[f]_a : Y \rightarrow \mathbb{R}$ and $[f]^b : X \rightarrow \mathbb{R}$ are defined by

$$[f]_a(y) = f(a, y) \text{ for } y \in Y \quad \text{and} \quad [f]^b = f(x, b) \text{ for } x \in X.$$

Theorem 1.46

Suppose $\mathcal{M}$ is a σ -algebra on X and $\mathcal{N}$ a σ -algebra on Y . Suppose $f : X \times Y \rightarrow \mathbb{R}$ is an $\mathcal{M} \otimes \mathcal{N}$ measurable function. Then

$[f]_a$ is an $\mathcal{M}$ measurable function on Y for every $a \in X$

and

$[f]^b$ is an $\mathcal{N}$ measurable function on X for every $b \in Y$.

Definition

Given a set X , a **Boolean algebra** (or simply, an **algebra**, when there is no risk of confusion) $\mathcal{A}$ on X consists of a collection of subsets satisfying:

- i) $X \in \mathcal{A}$.
- ii) $A \in \mathcal{A} \implies A^c = X \setminus A \in \mathcal{A}$.
- iii) If $A_1, A_2 \in \mathcal{A}$, then $A_1 \cup A_2 \in \mathcal{A}$.

Theorem 1.47

Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces. Then

- i) the set of finite unions of $\mathcal{M} \otimes \mathcal{N}$ -measurable sets is an algebra on $X \times Y$;*
- ii) every finite union of $\mathcal{M} \otimes \mathcal{N}$ -measurable sets can be written as a finite union of disjoint $\mathcal{M} \otimes \mathcal{N}$ -measurable sets.*

Definition

Suppose W is a set and $\mathcal{M}$ a collection of subsets of W . If for any increasing [respectively, decreasing] sequence (E_n) in $\mathcal{M}$ we have

$$\bigcup_{n \in \mathbb{N}} E_n \in \mathcal{M} \quad \left[\text{respectively, } \bigcap_{n \in \mathbb{N}} E_n \in \mathcal{M} \right].$$

Then $\mathcal{M}$ is called a **monotone class** on W . Moreover, the set

$$\mathcal{M}(\mathcal{C}) = \bigcap \{ \mathcal{M} \subseteq \mathcal{P}(W) : \mathcal{M} \text{ is a monotone class and } \mathcal{M} \supset \mathcal{C} \}$$

is called the **monotone class generated by $\mathcal{C}$** .

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Remark

Just as in the case for σ -algebras, $\mathcal{M}(\mathcal{C})$ is the smallest monotone class containing $\mathcal{C}$.

The Lebesgue Outer
Measure

Measurable Spaces

Measurable
Functions

Measure Spaces

The Lebesgue
Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class
Theorem

Product Spaces

Fubini's Theorem

Extension to
Complex-Valued
FunctionsLebesgue
Integration on $\mathbb{R}^n$

Theorem 1.48 (Monotone Class Theorem)

Suppose $\mathcal{A}$ is an algebra on W . Then $\sigma(\mathcal{A}) = \mathfrak{M}(\mathcal{A})$.

Theorem 1.49

Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ -finite measure spaces. If $E \in \mathcal{M} \otimes \mathcal{N}$, then

- i) $f(x) = \nu([E]_x)$ is a measurable function on X ;
- ii) $g(y) = \mu([E]^y)$ is a measurable function on Y .

Definition

*Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces and let $f : X \times Y \rightarrow \mathbb{R}$.
Then*

$$\int_X \int_Y d\nu(y) d\mu(x) f(x, y) \quad \text{means} \quad \int_X d\mu(x) \left(\int_Y d\nu(y) f(x, y) \right).$$

Example 1.50

Compute the following iterated integrals.

$$\int_{(0,4)} \int_{[0,4]} d\lambda(y) d\lambda(x) (x^2 + y) \quad \text{and} \quad \int_{[0,4]} \int_{(0,4)} d\lambda(x) d\lambda(y) (x^2 + y).$$

Lemma 1.51

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be two measurable spaces. Then for any $E \in \mathcal{M} \otimes \mathcal{N}$ and any $(x, y) \in X \times Y$ we have

$$\chi_E(x, y) = \chi_{[E]_x}(y)$$

and

$$\chi_E(x, y) = \chi_{[E]^y}(x).$$

Definition

Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ -finite measure spaces. Define the **product measure**, denoted by $\mu \otimes \nu$, as

$$(\mu \otimes \nu)(E) = \int_X \int_Y d\nu(y) d\mu(x) \chi_E(x, y)$$

for each $E \in \mathcal{M} \otimes \mathcal{N}$.

Example 1.52

Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ -finite measure spaces. If $A \in \mathcal{M}$ and $B \in \mathcal{N}$, then compute

$$(\mu \otimes \nu)(A \times B).$$

Proposition 1.53

Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ -finite measure spaces. Then $\mu \otimes \nu$ is a measure on $(X \times Y, \mathcal{M} \otimes \mathcal{N})$.

Theorem 1.54 (Tonelli's Theorem)

Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ -finite measure spaces and that $f : X \times Y \rightarrow [0, \infty]$ is a measurable function. Define

$$I_Y(x) = \int_Y d\nu(y) f(x, y), \quad I_X(y) = \int_X d\mu(x) f(x, y).$$

Then I_Y is an $\mathcal{M}$ -measurable function, I_X is an $\mathcal{N}$ -measurable function, and

$$\int_{X \times Y} d(\mu \otimes \nu) f = \int_X \int_Y d\nu(y) d\mu(x) f(x, y) = \int_Y \int_X d\mu(x) d\nu(y) f(x, y).$$

Corollary 1.55

If $\{x_{n,m} : n, m \in \mathbb{N}\}$ is a doubly indexed collection of nonnegative numbers, then

$$\sum_{n \in \mathbb{N}} \sum_{m \in \mathbb{N}} x_{n,m} = \sum_{m \in \mathbb{N}} \sum_{n \in \mathbb{N}} x_{n,m}.$$

Theorem 1.56 (Fubini's Theorem)

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces, let $f : X \times Y \rightarrow \overline{\mathbb{R}}$ be an $\mathcal{M} \otimes \mathcal{N}$ -measurable function, and suppose that

$$\int_{X \times Y} d(\mu \otimes \nu) |f| < \infty.$$

Then

$$\int_Y d\nu(y) |f(x, y)| < \infty \quad \mu\text{-a.e.} \quad \text{and} \quad \int_X d\mu(x) |f(x, y)| < \infty \quad \nu\text{-a.e.}$$

Furthermore, the functions

$$I_Y(x) = \int_Y d\nu(y) f(x, y) \quad \text{and} \quad I_X(y) = \int_X d\mu(x) f(x, y)$$

are $\mathcal{M}$ -measurable and $\mathcal{N}$ -measurable, respectively, and

$$\int_{X \times Y} d(\mu \otimes \nu) f = \int_X \int_Y d\nu(y) d\mu(x) f(x, y) = \int_Y \int_X d\mu(x) d\nu(y) f(x, y).$$

The Lebesgue Outer Measure

Measurable Spaces

Measurable Functions

Measure Spaces

The Lebesgue Measure

Simple Functions

Abstract Integration

Properties of $\int d\mu$

Cross Sections

Monotone Class Theorem

Product Spaces

Fubini's Theorem

Extension to Complex-Valued Functions

Lebesgue Integration on $\mathbb{R}^n$

Definition

Let $f : X \rightarrow \mathbb{C}$, then we say that f is **measurable** if both $\Re(f)$ and $\Im(f)$ are. Let f be a measurable function. We say that f is μ -integrable if

$$\int d\mu |f| < \infty.$$

In this case we set

$$\int d\mu f = \int d\mu \Re(f) + i \int d\mu \Im(f).$$

The space of all μ -integrable functions $f : X \rightarrow \mathbb{C}$ is denoted by $\mathcal{L}_{\mathbb{C}}^1(X, \mathcal{M}, \mu)$ but since we are going to be mostly working on such space, we will omit the symbol $\mathbb{C}$ and assume that every function is complex valued, if it is real-valued we will consider the same space but with the understanding that $\Im(f) = 0$.

[The Lebesgue Outer Measure](#)[Measurable Spaces](#)[Measurable Functions](#)[Measure Spaces](#)[The Lebesgue Measure](#)[Simple Functions](#)[Abstract Integration](#)[Properties of \$\int d\mu\$](#) [Cross Sections](#)[Monotone Class Theorem](#)[Product Spaces](#)[Fubini's Theorem](#)[Extension to Complex-Valued Functions](#)[Lebesgue Integration on \$\mathbb{R}^n\$](#)

Theorem 1.57

Let $(X, \mathcal{M})$ is a measurable space. If $f : X \rightarrow \mathbb{C}$ is measurable and $0 < p < \infty$, then $|f|^p$ is measurable.

Theorem 1.58 (Generalised Darboux's Inequality)

Suppose $(X, \mathcal{M}, \mu)$ is a measure space and let $f : X \rightarrow \mathbb{C}$ be μ -integrable.
Then

$$\left| \int f \right| \leq \int |f|.$$

Definition

Let (M, d) be any metric space. For any $x \in M$ and $r > 0$, the **(abstract) open ball of radius r around x** is the set

$$B_r^d(x) = \{y \in M : d(y, x) < r\},$$

and the **(abstract) closed ball of radius r around x** is

$$\overline{B}_r^d(x) = \{y \in M : d(y, x) \leq r\}.$$

Definition

The **supremum norm** on $\mathbb{R}^n$ is defined as

$$\|x\|_{\infty} = \sup\{|\pi^{\mu}(x)|\}_{\mu}$$

where π^{μ} is the μ -th canonical projection. It is a well-known fact that any norm induces a metric, the metric induced by $\|\cdot\|_{\infty}$ is denoted by d_{∞} . Similarly, abstract open balls with respect to this metric are written as $B_r^{\infty}(x)$.

Definition

The **open cube of side length 2δ centred at x** is defined as the abstract open ball $B_\delta^\infty(x)$.

Definition

We adopt the following common convention:

$$\mathbb{R}^m \times \mathbb{R}^n \text{ is identified with } \mathbb{R}^{m+n}.$$

In this sense,

$$B_\delta^\infty(x) \times B_\delta^\infty(y) = B_\delta^\infty(x, y)$$

for all $x \in \mathbb{R}^m, y \in \mathbb{R}^n$, and $\delta > 0$. Furthermore, the topology induced by $\|\cdot\|_\infty$ on $\mathbb{R}^n$ is denoted by $\mathcal{T}^n$ with $\mathcal{T}^1 \equiv \mathcal{T}$.

Theorem 1.59

Suppose that $G_1 \in \mathcal{T}^m, G_2 \in \mathcal{T}^n$. Then $G_1 \times G_2 \in \mathcal{T}^{n+m}$.

Notation

The Borel σ -algebra $\sigma(\mathcal{T}^n)$ on $\mathbb{R}^n$ is denoted by $\mathcal{B}^n$. Moreover, $\mathcal{B}^1 \equiv \mathcal{B}$.

Theorem 1.60

The collection

$$\mathcal{Q} = \{B_r^\infty(x) : r \in \mathbb{Q} \text{ and } x \in \mathbb{Q}^n\}$$

of rational balls is a basis for the topology $\mathcal{T}^n$ on $\mathbb{R}^n$.

Corollary 1.61

A subset of $\mathbb{R}^n$ is open if and only if it is the countable union of open cubes in $\mathbb{R}^n$. Furthermore, $\mathcal{B}^n$ is the smallest σ -algebra containing all open cubes in $\mathbb{R}^n$.

- The Lebesgue Outer Measure
- Measurable Spaces
- Measurable Functions
- Measure Spaces
- The Lebesgue Measure
- Simple Functions
- Abstract Integration
- Properties of $\int d\mu$
- Cross Sections
- Monotone Class Theorem
- Product Spaces
- Fubini's Theorem
- Extension to Complex-Valued Functions
- Lebesgue Integration on $\mathbb{R}^n$

Corollary 1.62

$\mathcal{B}^n$ is the smallest σ -algebra on $\mathbb{R}^n$ containing all the open cubes in $\mathbb{R}^n$.

Theorem 1.63

$\mathcal{B}^n \otimes \mathcal{B}^m = \mathcal{B}^{n+m}$. Consequently, $\otimes$ is associative, in the sense that

$$(\mathcal{B}^m \otimes \mathcal{B}^n) \otimes \mathcal{B}^p = \mathcal{B}^m \otimes (\mathcal{B}^n \otimes \mathcal{B}^p) = \mathcal{B}^{m+n+p}.$$

The Lebesgue Outer Measure
Measurable Spaces
Measurable Functions
Measure Spaces
The Lebesgue Measure
Simple Functions
Abstract Integration
Properties of $\int d\mu$
Cross Sections
Monotone Class Theorem
Product Spaces
Fubini's Theorem
Extension to Complex-Valued Functions
Lebesgue Integration on $\mathbb{R}^n$

Definition

The **Lebesgue measure** on $\mathbb{R}^n$ is denoted by λ^n and defined inductively by

$$\lambda^n = \lambda^{n-1} \otimes \lambda$$

with λ the Lebesgue measure on $(\mathbb{R}, \mathcal{B})$.

Theorem 1.64

Suppose $t > 0$. If $E \in \mathcal{B}^n$, then $tE \in \mathcal{B}^n$ and $\lambda^n(tE) = t^n \lambda^n(E)$.

Definition

The **open unit ball** is denoted by $\mathbb{B}^n$ and defined as the abstract open ball

$$B_1(0) = \{x \in \mathbb{R}^n : d(x, 0) < 1\}$$

with d the Euclidean metric.

Theorem 1.65 (Formula for the Hypervolume of the Unit Ball)

Let λ^n be the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Then

$$\lambda^n(\mathbb{B}^n) = \begin{cases} \frac{\pi^{n/2}}{(n/2)!} & n \text{ even,} \\ \frac{2^{(n+1)/2} \pi^{(n-1)/2}}{n!!} & n \text{ odd.} \end{cases}$$

Corollary 1.66

The hypervolume of a Euclidean ball of radius r is given by

$$\lambda^n(B_r(x)) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} r^n = \frac{1}{n} \lambda^{n-1}(\mathbb{S}^{n-1}) r^n,$$

where $\mathbb{S}^n$ is the unit n -sphere defined as

$$\mathbb{S}^n = \{x \in \mathbb{R}^n : \|x\| = 1\}$$

with $\|\cdot\|$ the Euclidean norm.

Theorem 1.67 (Clairaut-Schwarz Theorem)

Let G be an open subset of $\mathbb{R}^2$. Then

$$\partial_x \partial_y = \partial_y \partial_x$$

on $C^2(G)$.

Definition

A **multi-index** is an n -tuple $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ where each $\alpha_i \in \mathbb{N} \cup \{0\}$. The **order** (or **length**) of α , denoted by $|\alpha|$, is defined as

$$|\alpha| = \sum_{i=1}^n \alpha_i.$$

Definition

Let α be a multi-index and define the differential operator

$$D^\alpha = \begin{cases} \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} & |\alpha| \neq 0, \\ \text{Id} & |\alpha| = 0. \end{cases}$$

If $|\alpha| = m$, then we say that D^α is a **partial derivative of order m** .

A function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is said to be **class $C^m(\mathbb{R}^n)$** if all partial derivatives of order m exist and are continuous. A function is said to be **class $C^\infty(\mathbb{R}^n)$** (or **infinitely differentiable**) if it is class $C^m(\mathbb{R}^n)$ for all $m \in \mathbb{N}$.

Theorem 1.68

Let U be an open subset of $\mathbb{R}^n$ and let α, β be any two multi-indices with $|\alpha| = |\beta| \leq k$. Then

$$D^\alpha = D^\beta$$

on $C^k(U)$.